THE

JOURNAL-LANCET

The Journal of the Minnesota State Medical Association
and Official Organ of the
North Dakota and South Dakota State Medical Associations

PUBLISHED TWICE A MONTH

VOL. XXXV Minneapolis, March 1, 1915 No. 5

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Title: The Journal-Lancet, Vol. XXXV, No. 5, March 1, 1915

Author: Various

Editor: W. A. Jones

Release date: July 11, 2021 [eBook #65823]
Most recently updated: October 18, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/65823

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FEEDING OF THE HEALTHY INFANT [1]

By E. J. Huenekens, A. B., M. D.
Instructor in Pediatrics, University of Minnesota
MINNEAPOLIS

[1] Read before the Hennepin County Medical Society, Nov. 2, 1914.

The science of infant-feeding has been revolutionized in the last twenty years, and, in the process, it has advanced too radically in many directions. Lately, the pendulum has been swinging backward, so that the most advanced knowledge of today probably represents a middle ground between extreme radicalism and extreme conservatism. In no other direction is this more manifest than in the feeding intervals. The religious adherence to the four-hour feeding interval is giving way to a more rational system. I am one of the firmest adherents of the longer interval: the food is better digested, the stomach has a period of rest, and the general well-being of the infant is better furthered than with more frequent feedings. But there are certain infants who do not receive enough nourishment in this interval, especially young breast-fed infants in whom it can be demonstrated by accurate weighing, before and after nursing, that they receive considerably more milk in twenty-four hours with the three-hour interval. This is the more important in that Rosenstern has demonstrated that a large proportion of infants up to the age of six weeks require more than the usual 100 calories per kilogram of body-weight. One hundred calories represents 150 grams of breast-milk, so that a five-kilo, or eleven-pound, baby should receive a minimum of 750 c.c. of breast-milk in twenty-four hours.

By far the best food for the healthy infant in every way—and this cannot be emphasized too strongly—is mother’s milk. There are certain alimentary disturbances in which it may be advisable to replace breast-milk with certain artificially prepared foods, such, for instance, as albumin milk in alimentary intoxication; but this is never true of the normally healthy infant. While, as regards growth and freedom from digestive disturbances, certain artificially prepared foods may, when used with exceeding care, produce as good results as breast-milk; nevertheless, this is only one function of breast-milk. The other function which can be imparted to no artificial food is the passive immunization of the child against infection. Ehrlich (*Zeit. f. Hyg. u. Infectionskr.*, 1892, xii, 183) has proved that antibodies, antitoxin, and agglutinins are transmitted directly through the milk from mother to child; and it has been shown that the blood of a breast-fed child is considerably more bactericidal than the blood of a bottle-fed infant.

The practice of weaning the baby for trivial reasons has increased in the last decade, and can be laid largely at the door of the medical profession. For all practical purposes the only absolute indication for weaning the baby is open tuberculosis in the mother. For the last few years I have been making a systematic inquiry at the University Dispensary and Infant Welfare Stations as to reasons for weaning young infants; and in nine cases out of ten, the answer has been that "the milk gave out." In only a very small proportion of cases has an ordinarily well-nourished mother insufficient milk; far oftener the fault lies with the child. Insufficient and late development of the sucking reflex prevents these infants from completely emptying the breast, which in time "dries up." This period can be tided over by nursing from both breasts, by temporarily increasing the number of nursings, or temporarily employing "allaitement mixte." In cases in which, after long, patient effort the supply of milk is still insufficient, either supplementary or complementary feeding of cow's milk can be given. Where this mixed feeding is employed a minimum amount of cow's milk should be given; and the opening in the nipple should be as small as possible, otherwise the child gets too much cow's milk, and with too little effort, and gradually refuses the breast.

Another excuse, and one fostered to some extent by physicians, is, that certain breast-milks are "poison for the baby." This has even less foundation in fact; and here again the fault must be looked for in the baby rather than in the mother. Outside of certain variations in the fat-content, all breast-milks are alike in composition. In proof of this Finkelstein has fed these babies at the breast of tried wet-nurses with absolutely no benefit, while the children of the wet-nurses would thrive at the breast of the "poison-milk mother."

Abscess of both breasts may force a temporary cessation of nursing, but the breast should be regularly emptied until the inflammation has subsided; and then the nursing should be re-established. Cracked or sunken nipples may render nursing impossible, but they do not stop the flow of milk. In both these latter conditions the milk may be manually expressed or removed with the breast-pump. In this connection I wish to recommend the improved Jaschke pump, in which, by means of a releasing valve, the sucking movements of the child can be very closely imitated.

Where artificial feeding must be started early, cow's milk is almost universally employed. Whenever possible, "certified milk" should be used; the ordinary milk, however, can be boiled with little or no harm. In diluting and preparing this milk, we have the choice of several methods. The so-called percentage feeding, favored in America, is difficult and cumbersome, and has no advantages over its simpler rivals. Pfaundler's rule may be safely employed. It is as follows: One-tenth body-weight of milk, one one-hundredth body-weight of sugar diluted up to one liter; give 200 c.c. five times in twenty-four hours. Even simpler is the following: One-third milk for the first month, one-half for the second month, two-thirds for the third and fourth months, each with the addition of 4 to 6 per cent sugar. Either milk-sugar or ordinary granulated sugar may be employed. The malt sugars and extracts should be reserved for sick children. After the second month, oatmeal water may be used as a diluent in place of plain water.

Recently Friedenthal, a Berlin physiologist, has attempted an exact imitation of mother's milk, including that important element, the salt, which had, until recently, been entirely neglected. Langstein is very enthusiastic over this milk as a food for healthy infants; but Finkelstein, in a personal communication, assured me that it has not as yet proved itself. Schloss, dissatisfied with the results of the Friedenthal milk, has modified it in the direction of casein milk by replacing the milk-sugar with the malt preparations, and increasing the protein content. He claims good results, and is supported by Leopold, of New York, who has used it extensively. But we must leave the final word as to both these milks for the future to decide. From the sixth to the ninth month for both breast-fed and bottle-fed babies, cooked cereals, toast, and vegetables should be gradually added to the diet. At the ninth month, unless this is one of the hot summer months, the nursling should be weaned, and a small amount of cow's milk substituted. The weaning should be gradual by omitting one nursing period each week. The one important exception to the foregoing rules for the first year of life, is the premature infant. In the ninth month of fetal life, reserves of calcium and iron are stored up in the body, which the infant gradually uses up during the first nine months of extra-uterine life. The premature infant lacks this store, and manifests it in different ways. As early as the second or third month a breast-fed premature infant may develop a most extensive craniotabes. This is not due to a true rachitis, i.e., disturbance of calcium metabolism, but to a want of calcium in the body. Small amounts of cow's milk, which contains much more calcium than human milk, or calcium in the form of calcium lactate or chloride, will remedy this condition. A similar process happens in the case of iron. The premature infant is born with a hemoglobin percentage of 100 to 110; by the third or fourth month this may sink to 40 per cent, and for this reason green vegetables should be added to the diet as early as the fourth month.

The diet of healthy children in the second year should include cooked cereals, vegetables, toast, cooked fruits, and meat-juices; and the quantity of cow's milk should be limited to one and one-half pints in twenty-four hours. The question of the addition of meat to the diet is important. Some authors have recently advocated the giving of meat as early as the ninth month. During the past year, working in Finkelstein's laboratory, I have been able to gather some facts which have a direct bearing on this question. (*Zeitschrift für Kinderheilkunde*, July, 1914.) By means of the new electrometric determination of absolute acidity (that is, the number of H ions), I was able to show that the acidity of the stomach before the eighteenth month of life is insufficient to permit any peptic, i. e., protein, digestion. Solomon, working in the same clinic, in a report not yet published, has shown the same thing from a clinical standpoint. He found that on a meat diet up to the end of the second year large quantities of muscle fibers passed through with the bowel-movement unchanged; but after that age they rapidly decreased in number. It is, therefore, worse than useless to add meat to the diet before the beginning of the third year.

Eggs frequently produce profound disturbances in young infants, perhaps on account of the absorption of egg albumin, unchanged, in the blood-stream; and they should be kept from the diet-list until the beginning of the fourth year.

These rules for feeding are generalized, and there may be many exceptions. Each child is to some extent a law unto itself, and this is especially true of those children with nervous or exudative diathesis.

In conclusion, I wish to make a plea for greater uniformity in our rules for infant-feeding. Even more than in strictly medical affairs has the public the right to demand information. Heretofore, every new book and every public lecture on infant-feeding has deviated markedly from its predecessors, until the confused laity, and even general practitioners, have turned in disgust to proprietary foods and formulas. Pediatrics is a new science, and as such is bound to undergo rapid changes and conflicting opinions. But that need not hinder us from agreeing on certain fundamental facts which can be given as guides to the general practitioner and to the public.

I believe that the simple rules for infant-feeding here laid down are neither too conservative nor too radical to serve as a basis of agreement upon which the medical profession may show to the public a united front on this important question. Such uniformity of opinion—and the sooner it can be reached the better—will not fail to have a beneficial effect on both the profession and the public.

DISCUSSION

Dr. Jacob Hvoslof: I would like to ask about the value of lime-water added to the milk. I recently had an experience where I mixed an ounce of lime-water to a pint of milk, as I thought that would improve it. but for some reason or other the baby would not digest his milk. After a while I left the lime-water out, and everything went well. Whether this is a “post” or “propter” I should like to find out.

Dr. O. R. Bryant: In case of an exudative diathesis, where you probably will start solids early, you will also be able to use meat earlier. An infant that does well on solids at six months can probably have meat once a day at fifteen months and show a normal stool.

Dr. S. R. Maxeiner: I would like to ask Dr. Huenekens where he classes eggs and egg albumin.

Dr. C. G. Weston: I have been very much interested in Dr. Hueneken’s paper. I care only for the babies during the three or four weeks after birth; and of late years many of them have passed from me directly into the hands of the pediatrists. I formerly had the babies nursed every three hours, but finding that the baby specialist immediately, on assuming charge, put them on the four-hour schedule, I changed, about a year and a half ago, to that interval; and I thought my troubles would cease, but such has not been the case, and it has been my impression, as well as that of the nurses who have had the care of the infants, that it has made very little difference.

The four-hour schedule is not a new thing in Minneapolis. Many of the older members of this Society may remember that twenty years ago Dr. R. O. Beard always fed his babies in this way.

It seems to me that we should make no hard and fast rules for the feeding of babies, except the one that mother's milk should be used whenever possible. We should individualize with the babies. If they do well on the four-hour schedule, follow it, as it makes the care of the child easier for the mother; if, however, the child does not get sufficient milk on this interval to properly nourish it, diminish the latter to three hours.

The only way to accurately determine how much milk the nursing infant is getting, is to weigh the baby before and after nursing. One is often surprised at the varying amounts obtained by the same baby at different nursings with no obvious difference in the condition of the breasts. We have had a baby obtain as much as three ounces in the first five minutes of nursing, and at the next feeding take only one or one and a half ounces in twenty minutes.

The green and frequent stools, with evidences of colic, etc., are often found to be due to too much milk, or taking it too rapidly; and the weighing method is the only way to determine this.

I most heartily endorse what Dr. Huenekens said with reference to the importance of encouraging in every way maternal nursing. Many a mother gives up the attempt to nurse her baby on account of some soreness of the nipples or because she has thought she had too little milk to be of any use. Most of these cases may become, by the means recommended by the reader, good milkers, and many a baby's life may thus be saved.

Dr. E. K. Green: I would like to ask a question in regard to putting babies on cow's milk. I have adhered very closely to the principle that modified cow's milk is absolutely the best food for infants, if it is impossible to get mother's milk, but many times when I have had the opportunity to follow these cases carefully I have had all sorts of stomach and bowel disturbances on cow's milk until someone would suggest some other food, such as malted milk, or Mellin's Food, or even condensed milk, which seems to be the farthest from the natural food, and then the babies would get along fairly well. Is this a common experience, or is there something wrong with my method? We have in our own home two children brought up on the bottle, one with malted milk and the other with Mellin's Food. In both these cases I tried, not only once, but several times to use the modified cow's milk, but failed absolutely. I would like to know if you consider the fault usually with the modified milk, or does the individual have considerable to do with the case?

Dr. A. S. Fleming: I would like to ask if in the case of the healthy infant the mother's diet would modify the constituents of the milk otherwise than in the facts stated. For instance, will it modify the character of or the percentage of the sugar, or will any of the aromatic constituents disturb the infant's digestion?

Dr. M. J. Jensen: Dr. Huenekens dealt with the feeding of the healthy infant only. I would like to ask if it is not true that nearly all infants born alive, are born as healthy and sound as any infant ever is, so far as the functions of its organs and tissues are concerned? Nature frequently decides on producing premature births and "still"-births, rather than running the risk of producing a sick or sickly infant. In young infants it is very often difficult to determine when to classify them as healthy or unhealthy, realizing the conditions of their environment and usual care that is given in the homes.

In regard to the sterilization or boiling of cow's milk: I do not think that children fed on pasteurized or boiled milk develop as well as those who are fed upon raw milk as it comes from the cow. Dr. Palmer, of Chicago, fed seven hundred children on raw milk during the midsummer months and only lost three of the number. The miserable, atrophied children began to live the moment treatment with raw milk was begun. If the process of milking was carried out in a sanitary manner, or by means of a suction apparatus, then cooled, and placed in sterilized bottles, I believe we would prohibit the development of bacteria, and save the food which exerts so marked a protective influence upon the infant's organs.

When raw milk free of all objections cannot be obtained, it is sometimes advisable to use another milk product namely, buttermilk.

DR. S. MARX WHITE: There is just one point I have been thinking about in the discussion on the question of infant-feeding, and that is whether Dr. Huenekens really means us to believe that in practically all cases the mother can furnish sufficient milk for the child. He passed that over in saying that in nine out of ten cases the mother gave as a reason for discontinuing the milk that the milk gave out. Is it not true that in a good many instances the mother needs treatment quite as much as the infant? I do not mean medical treatment, but management. Is it not true that an overworked, tired, nervous, worrying mother is unable to supply sufficient milk for the child? It has been my impression from a very limited experience in this field, that the mental and nervous and physical state of the mother is a very large factor in the production of the milk. When upset and under deleterious influences she is really not a proper producer for the child; and the management of the mother is often quite as important a factor as any other.

DR. W. H. AURAND: In such cases as Dr. White just mentioned, what are we going to do to increase the supply of milk? Also, I would like to ask Dr. Huenekens if he means to feed to the new-born baby 200 c.c. at a feeding?

DR. HUENEKENS (closing): As regards lime-water: I cannot recommend its use. Wherever there is a specific demand for calcium, as in premature infants or spasmophilic cases; or where it may help to produce a firm stool; or, as in diarrheal disturbances, it may be of great value, but in the normal healthy infant it is of no benefit whatever.

Dr. Bryant mentioned the giving of meat in exudative diathesis: His statement that such infants can probably have meat once a day at fifteen months, and show normal stools, is beside the question. A normal macroscopic stool does not necessarily mean that the meat has been digested. However, I am now working on this problem, that is, to determine whether an early solid diet produces an earlier digestion of meat.

I would classify eggs and egg albumin as proteins, and therefore not digestible until the beginning of the third year: but, over and above this, there is danger of anaphylaxis from the absorption of the unchanged egg albumin into the blood-stream.

What Dr. Weston says of the feeding intervals is very interesting. I do not want to be considered an enemy of the four-hour feeding, for I use it wherever possible, and I think it the best interval; but when the infant cannot get enough in that period, we have to choose between two evils. I think the lesser evil is to give the child more milk at shorter intervals, and take the risk of a slightly poorer digestion. We should, also, wherever possible, control the amount of breast-milk by weighing the child before and after nursing. It is highly important to determine whether the baby is getting too much or too little.

As to Dr. Green's statement, "Modified milk" is a very general term. What is usually meant is milk with a high percentage of fat and a low percentage of sugar, while malted and condensed milk have a high percentage of carbohydrate. In my opinion, if he had used cow's milk without the addition of cream and with large amounts of cane sugar, he would not have had this trouble. But a large number of children will not do well on this diet. We have special rules for abnormal children with exudative and nervous diathesis.

In reply to Dr. Fleming’s question regarding the mother’s diet and its effect on her milk: What the mother eats has absolutely no effect on the composition of the milk in any way whatever, except perhaps in the percentage of fat. Now-a-days we do not advise any particular foods for the mother’s diet,—anything she likes, and can digest, plus large quantities of fluid;—otherwise there is no single food we advocate—none that will make the milk richer or better, or increase the quantity.

I cannot agree with Dr. Jensen that raw milk is so far superior to boiled milk. Of course, wherever it is possible, we should use certified milk, which does not require boiling; but, if we have inferior cow’s milk contaminated with bacteria, we can boil the milk with very little harm. It is just as well digested, and the food value just as great. There is of course slight danger of scurvy; but that is very easily diagnosed, and very easily cured by a little fresh milk or small doses of orange juice. Where we have inferior milk, it should be boiled in every case.

Dr. White brought up a very interesting point in regard to nervous mothers. Their milk supply is subject to wide fluctuations; but, if the breasts are well emptied at each nursing, they will secrete sufficient milk. I will admit that these cases are difficult to handle, for the infants usually have nervous diathesis, and do not respond well to ordinary food. The one important point is to completely empty the breasts; and that is the only measure we can take to increase the supply of milk.

In reply to Dr. Aurand: I would feed a new-born infant 200 c.c. at a feeding if the milk is sufficiently diluted. The liquid part of the food passes very quickly into the duodenum, so that, before the infant has finished feeding, a part of this quantity has already left the stomach.

In conclusion: We have an opportunity in our infant-feeding to practice the really scientific prophylactic medicine of the future. We can do more in preventing infant-mortality by proper feeding than by any other single measure; and we should encourage mothers to bring their new-born infants to the physician for advice on feeding, and to continue to consult him at longer or shorter intervals during the whole of the first year of life.

THE INEBRIATE [2]

By George H. Freeman, M. D.
Superintendent of the Minnesota State Hospital for Inebriates
WILLMAR, MINNESOTA

[2] Read at the 46th annual meeting of the Minnesota State Medical Association, St. Paul, October 1 and 2, 1914.

The Minnesota Legislature of 1907 passed a bill establishing the Hospital Farm for Inebriates, placing its management under the State Board of Control, and providing for its maintenance by setting aside 2 per cent of the saloon-license money for that purpose. Later, a law was enacted providing for the issuance of certificates of indebtedness; and active construction work soon commenced. The Hospital was opened on Dec. 26, 1912, with Dr. Tomlinson, formerly Superintendent of the St. Peter State Hospital, at its head. Through his untimely death, five months later, Minnesota lost one of her most faithful officials. The principles underlying the work at Willmar, are, with but slight change, those that he so earnestly advocated.

This paper is based upon the study of the patients admitted from the opening of the Hospital until the close of the biennial period, on July 31, 1914,—approximately eighteen months.

Patients are admitted to the Hospital following an examination in a probate court. In such cases there is no expense to the patient's relatives, except that they are expected to furnish clothing, and a little money for the purchase of tobacco and small luxuries. Voluntary patients are also received following their own application in a probate court. They pay at the rate of $1.00 a day, each month in advance. No distinction is made in the treatment of the two classes of patients, except that a voluntary patient cannot be detained if he wishes to leave. Any resident of Minnesota who is habitually addicted to the use of alcohol, morphine, cocaine, or other narcotics, may be admitted to the institution, provided the history of the patient, as furnished by a probate court, indicates that the man can be benefited by treatment. It is presumed that anyone can be benefited who wants to be, unless afflicted with irremediable chronic disease.

The requirement that the history be furnished, and the ability to refuse admission, have kept out of the Hospital many undesirable individuals who could be cared for only under the discipline of a well-regulated reformatory. However, some, no matter how carefully the history is taken, slip by. The majority of those discharged as not proper subjects, come from that class. As there are no accommodations for individuals suffering with tuberculosis, no one known to be suffering with that disease is admitted. Once in a while a tuberculosis individual gains admittance, but, if not too ill to be released, he is discharged.

During the eighteen-month period, 209 men and 32 women were regularly committed; and 18 men and 3 women were received as voluntary patients. In addition to those classified as voluntary patients, a considerable number have, of their own volition, applied for treatment, and, being unable to pay, have submitted to commitment, in order to obtain treatment for their habit.

There has been a fairly uniform increase in the number of patients received each month, which is gratifying, as showing the need of such an institution and also as an appreciation of the benefit that may be expected. During the last month of the period, twenty-five patients were admitted.

While the causes of inebriety are diverse, it is a significant fact that 182 patients, out of 262, assign associates as their reason for drinking; and observation of their history clearly shows that they have drifted along, drinking now and then and more and more each year. A few assign illness, domestic trouble, or financial worry as a cause for drinking. In only 6 instances was heredity noted. In 132 cases the parents were abstainers.

We have found it impossible to formulate any system of classification of the unfortunates under our care. In order that some idea may be obtained as to the number using alcohol and the various drugs, we have constructed the following table:

FORMS OF INEBRIETY

	Men	Women
Steady drinkers	130	2
Periodical drinkers	76	8

Morphinism	3	11
Alcohol-morphine	5	2
Alcohol-cocaine	3	..
Alcohol-heroin	3	..
Alcohol-morphine-cocaine	3	1
Alcohol-morphine-cocaine-heroin	1	..
Alcohol-morphine-veronal	1	..
Morphine-cocaine	..	1
Morphine-cocaine-heroin	2	..
Total	227	35

The treatment of the inebriate naturally divides itself into two stages: the treatment, first, of the immediate effects of indulgence, and, second, such treatment as will tend to prevent a repetition of the indulgence. The treatment of the immediate effect of alcoholic indulgence is regarded as the easiest part of the work. While patients are at times received under the influence of intoxicants, in no case have they been unruly. For an obstreperous intoxicated person the quickest soberer is apomorphine judiciously used; but we have never yet resorted to it. Generally, a fairly rapid reduction in the amount of alcohol consumed is made, instead of immediate withdrawal. Only in the most exceptional cases is alcohol given over three or four days. As a rule, during the first day it is given fairly freely. The treatment received during this period depends entirely upon the individual; and the treatment of one may be entirely different from that of another. Many receive baths at a temperature of 98° to 100° F. for thirty or sixty minutes for nervousness and sleeplessness. Some receive the coal-tar hypnotics, veronal or sulphonal; the more restless, hyoscine; and for others paraldehyde is used,—and occasionally chloral is used in combination with hyoscine and cannabis indica.

As long as he receives alcohol, the patient remains in bed and receives only liquid diet. In cases of considerable digestive disturbance, capsicum is freely used, but we have seldom found it necessary.

All receive preliminary catharsis, but no attempt is made at prolonged elimination in that way.

For about a month tonic treatment with strychnine nitrate is used in doses of 1-20 to 1-40 gr. three times a day. Any other medication depends entirely upon the physical condition of the patient as revealed on examination. Only under the most exceptional circumstances are drugs given in alcoholic vehicle.

In morphine or cocaine users, the reduction is usually made more gradually, requiring a week to ten days. Generally, we find a patient comfortable with one-half the drug he has been accustomed to taking. In some cases we find it best to reduce the quantity to about one-half grain, and then abruptly cease.

Under this plan, diarrhea, cramps, restlessness, and insomnia are much less marked. We regard the free use of the prolonged warm bath as more advantageous to those addicted to drugs than to alcohol. Generally, it is the only measure that seems to offer relief. We particularly do not use hypodermic medication in any drug users.

Heroin users, who seemingly are more numerous, receive their drug only once in twenty-four hours. The withdrawal of the drug does not cause the discomfort that the withdrawal of morphine causes. Vague sensations of discomfort, some perspiration, and insomnia are met with in such cases.

No users of cocaine only have been met with, but in mixed forms this drug is at once withdrawn.

The removal of alcohol or drugs is the easiest part of the work. Under the regular discipline of the institution, and the absence of temptation, the great majority of patients get along without any trouble because of abstinence. But there is the future to fear. The patient must go out into the world again, and engage in the daily struggle for his livelihood. One must aim to put him in such condition that he may be able to resist the temptations that will surround him on every hand. Our work, then, is to build up and re-educate, to strive to form a new character, to encourage a habit of sobriety, instead of drunkenness, to teach the man to work, to occupy himself, to obtain for him a new outlook on life, and to teach him his duty to himself, to his family, and to his neighbor. Here is where our difficulty begins. Nearly every inebriate has a firm belief in his ability to abstain from alcohol or drugs at any time and under any condition, because he thinks he is not really responsible for the condition into which he has fallen, and that, had not certain things happened, he would not have been drinking.

He is insistent in iterating and reiterating that he has now made up his mind to stop drinking, and that is all that is necessary. Though admitting that, for five, ten, or fifteen years, he has been going steadily downward, he has full confidence in himself, and he believes injustice is being done him when his parole is refused and he learns that he is expected to remain until he has strength to resist temptation.

In this upbuilding of body and character the following are essential: regularity of habits, discipline, work, food, and recreation, together with the personal influence of the physician and those coming into close and personal contact with the patient.

Regular work is one of the most valuable of the remedial agents at our command. It should be suited to the individual, and, as a rule, should not be that to which the man has been accustomed. Particularly is this true of the man who is used to mental labor only,—the clerk, the physician, the pharmacist, the merchant, etc. For them out-of-door work on the farm, lawn, or in the garden, is the very best, and next comes indoor shop-work. We must provide something that engages time and attention, that provides some new outlook upon life, and enlarges some field of endeavor in which the patient has labored before coming to the Hospital. Thus far the work has been on the farm or the improvement of the grounds, or has been carpenter, cement, or some construction work. The women do all the mending, and make all needed articles, such as bedding, towels, etc. They also work in the laundry. At present we are teaching embroidery of various kinds, no one of our patients having ever learned any such work.

The future must see us provided with shops, especially for winter work. With a capacity of ninety-nine men we are able to keep them fairly well occupied during the winter months, but any increase will have to be cared for under special conditions.

A very important factor is the length of time, as mentioned above under prognosis, that a patient remains under care. As a general thing, it is expected that the average patient will remain, approximately, six months.

The period of detention is determined only after a study of the individual. An endeavor is made to consider all factors that may influence the future life of the patient,—the length of time and the amount he has been drinking, the effect on his character and physical health, the surroundings and occupation to which he must return. Some patients are paroled at the end of six months, some remain seven months. Drug users require treatment for a much longer period of time than users of liquor; and they remain from nine months to a year. The law provides that a patient shall not be paroled in less than two months, nor shall he be detained longer than two years without parole. This, of course, introduces the disagreeable aspect of the work. The detention is compulsory; and in some patients antagonism possibly over-balances the benefit of detention.

“One of the most pronounced features of inebriety is, however, the inability of many inebriates to appreciate the necessity for treatment; and the more severe the inebriety, the less easy it is first to get the patient under treatment at all, and, secondly, to get him to remain long enough for any treatment to have a permanent curative effect. One has only to work among inebriates, no matter to what class of society they belong, to know that fear of interfering with the liberty of a subject who has no real liberty, in that he is a slave habitually or periodically to the drink craze, results in the interference with the liberty of all those who have to put up with his irresponsible behavior under the influence of alcohol and other narcotic drugs.

“Were the treatment of the inebriate only possible in a free sanatorium, only a small minority of inebriates would come under treatment at all, and these would be of the less severe type.” (Pathological Inebriety, by J. W. Ashley Cooper, 1913.)

Discipline is of great importance, but great care must be taken in its enforcement. It is of more value for one to perform a certain duty because one regards it either as the proper thing or as likely to benefit one’s self or others.

The personal influence of those who come into close contact with the inebriate can hardly be overestimated. He is easily influenced, often easily led, and a few thoughtless words or careless actions can undo the result of patient work.

All factors that may influence the future life of the patient must be taken into consideration,—the length of time and amount he has been drinking, the effect on his physical health and character, and the surroundings and occupation to which he must return. Very often the cause of the commencement of the patient's excessive drinking may be removed or may have disappeared. Such would favorably influence the prognosis.

The presence or absence of irremediable disease is important. For instance, a woman recently committed to our care suffered from what was supposed to be, or was, neuralgia. She still has occasional twinges of pain; but we believe when the dentist has finished his work these will disappear, and her prospect be reasonably bright. A man, 56 years of age, four years ago, suffering from stone in the bladder, was given morphine, following an operation. The bladder condition was permanently relieved, but he became a morphine user. Such a case is a promising one. In him the destruction of character is but little marked.

A boy, chasing around the city, acquired the cocaine habit, and became a loafer, drinker, and follower of loose women. For him the future offers practically no prospect. Were he a little younger, and had the attempt to rescue him been made earlier, there would have been much more promise. But I doubt whether he can withstand the lure of his former life. With a few drinks, his judgment becomes paralyzed, and he is back to cocaine again.

Another man, an alcoholic, a printer, became nervous and exhausted after six months of linotype work. He probably will not get over his drinking permanently unless he changes his occupation.

One of the most important factors as regards recovery is the length of time a patient remains at the Hospital. It is sheer folly to expect that in a few short weeks a man shall have entirely recovered from the effect of excesses extending over a period of years, to expect him to regain a lost will power in that time.

Another important factor is the insight a patient obtains into his own condition. We cannot claim to make a man stop drinking. All we can do, is to place him in such mental and physical health that it is unnecessary for him to resort to stimulants.

RESULTS

The result of treatment in a disease of the nature of inebriety, can hardly be estimated in such time as the Hospital has been open. Our statistics are simply offered to show possibilities. As the statute under which the Hospital operates, contemplates a period of detention and treatment for not less than two months,—and that period is even too short in the vast majority of cases,—anyone resident in the Hospital for less than two months has been placed in a separate class, and we can learn that only two of these are doing well. Of 172 men, aside from those who have been discharged as not proper subjects, 54 were paroled, of whom 37, or 68 per cent, are reported as doing well, 27 were released under bond, of whom 17, or 63 per cent, are reporting. Over one-half of the voluntary patients are reporting.

Averaging all, we find 57 reporting as doing well; 30 fail to report; 29 are escaped, and we can learn nothing of them; and 56 were here less than two months, 38 of these being escaped; 7 voluntary patients; and 8 were released under bond. A percentage of abstainers of 25, is to be regarded as most excellent; and as one-third of those who have left here are still abstaining, the greater number of failures occurring in the first month, the outlook for the future is very encouraging.

So far, we have been speaking of what we are trying to do for the more hopeful class of patients. But what are we to do in the future with the incurable, the recidivists? Are we to send them back into the world time and again, let them abuse themselves, perchance their families, and let them be, as it were, a constant menace to society? No, society has a right to protect itself and to protect an individual against himself. There should be provision made for this class. They should be cared for in an institution under strict discipline, and made to support themselves there and to contribute to the support of those who may be dependent upon them.

As soon as considerable numbers are received at an institution, the more apparent becomes the need of means for classification, especially as to character. It is decidedly unwise to allow the intermingling of the young lad who has just commenced to drink, with the incorrigible or the sodden, whose every thought may lie bestial.

The most practical means of classification is by the use of cottages; and it is on that plan that Minnesota’s institution has been started. If two cottages were built at Willmar we should be able to make four groups of patients with decided advantage to our inmates. Not more than forty inmates should be cared for in each cottage: and I am strongly in favor of separate rooms for sleeping-quarters, instead of dormitories.

SUMMARY

The essential in the treatment of the inebriate as we meet him, is upbuilding of body and character, which requires time, and in which drugs play only a small part.

Compulsory abstinence is of great value if we expect to care for a majority of the inebriates.

It would be wise for the State to undertake the custody, care and control of all non-criminal inebriates in one institution, provided adequate facilities for classification were available.

DISCUSSION

Dr. C. R. Ball (St. Paul): I have been very much interested this afternoon in this symposium on the treatment of fractures, the last word in obstetrics, and the inebriate, only it seems to me the Program Committee put the cart before the horse, and should have put the inebriate first, and the other things would naturally follow afterwards.

Dr. Freeman has splendidly presented his work and results at the Willmar institution. It is a subject to which I think medical men pay too little attention. I have more and more come to look upon the inebriate as a type of nervous disease and, in the great majority of cases, a functional nervous disease. It may be classified as we classify nervous diseases. We classify in one way functional nervous diseases as to their cause,—acquired, hereditary and acquired, or wholly hereditary.

The inebriate may be also classed in the same way. There are perhaps a few cases in which the habit of taking alcohol is absolutely acquired, but they are comparatively few. There are also a few cases of nervous prostration or functional nervous conditions from overwork, from a depleted condition, where the nervous condition comes on; and we may say it is acquired, and the prognosis in both of these cases is good. It requires but little effort to put them on their feet. Then we have that larger class of neurasthenic or functional nervous conditions, belonging to the second group, in which the nervous disease, as well as the inebriety, is partially acquired and partially hereditary. There is a large class here. They have an unstable nervous system, and whether they drink or break down depends a great deal upon the environment and physical condition. This type of inebriate must be treated along the same broad lines that we treat a person who is a neurasthenic, who is subject to repeated nervous breakdowns.

There is another type which, unfortunately, is rather large; and this is the wholly hereditary, and in this type we may classify the dipsomaniac. I have looked for a long time upon dipsomania as a periodical nervous disturbance, similar to periodical attacks of migraine or epilepsy, or periodical attacks of insanity. Often where a son is an inebriate we find a history of migraine in the mother. Very often there is insanity, and very often there is epilepsy, so that when we come to consider the dipsomaniac we have a tremendous problem. He does not drink for the fun of it, but chiefly because of mental depression, mental restlessness, which is so great that he turns to alcohol to buoy up his spirits and get rid of the feeling which rather than suffer with, he would often prefer to die. I have a man of that description who came to me, and said that at a certain time he became depressed and suspicious, began to hate himself, went along the back streets, absented himself from his usual associates, and always did this at the beginning of his drinking bout. That is the case with all dipsomaniacs. It is a disease similar to epilepsy, and our success in treating this type is just about as good as in treating epilepsy. It is not the alcohol: it is an inherited condition; it is a periodical nervous disturbance, just as epilepsy and migraine are.

We hear a great deal about the prevention of tuberculosis, and much is done to prevent it. I think we hear much more about the evil effects of syphilis than of alcohol, but, in my experience, I would place alcohol at the top of the list as being the most damaging both to the individual himself and to his offspring. We have heard a great deal about the effect on the offspring. In my clinic at the Free Dispensary I have many epileptic children, and I should say in sixty per cent of the cases one parent is an alcoholic. An address of Dr. Rogers, of Faribault, with reference to the ill effects of one intoxication, when a conception occurs during that time, put the subject of drinking before me in a new light. Much interesting experimentation has within recent years been done with rabbits and guinea-pigs to show the harmful effect of a single dose of alcohol given to either the male or female parent before conception, on the after-coming litter.

Not long ago I read an article by some man in New York in which he stated he had traced seven cases of epilepsy to the evil results of a single intoxication in seven different parents. That was something rather new to me, as I thought, in order to get the bad effects, on the descendants, of alcohol, it was necessary to be a chronic alcoholic, and I believe very few of the laity understand that, if conception happens to occur during one drunk, the parent being otherwise a temperate person, the ill effects may be visited on the offspring to as great an extent as if the parent were a chronic inebriate. These are some of the things which would do good if given publicity.

In regard to the treatment: I can fully agree with Dr. Freeman in everything he has said. There is certainly no specific when you come to consider the nature of the trouble. The treatment must be carried along the same general lines of physical and moral upbuilding as those we seek to follow in functional nervous disease.

Dr. W. A. Jones (Minneapolis): I wonder how many members of the State Association have visited the hospital for inebriates at Willmar. I would like to ask all those who have, to hold up their hands. Five or six of this audience, representing the twelve hundred doctors belonging to this Association. That gives one a fair estimate of those familiar with the State farm for inebriates.

I should like to know further how many members of the legislature have visited this institution, and how many have tried to condemn it or perhaps to take it for a tuberculosis hospital. That is what they will do unless we physicians stand by Dr. Freeman and the institution.

There is too much sentiment, too much sympathy among friends, relatives, courts, juries, and charity workers, as to the inebriate; but once he gets to Willmar and is under a proper regimen, his attitude changes totally toward himself and toward the world. After one has watched the treatment at Willmar and has seen the benefit these patients derive, he wonders why so many women and so many men are sent to quack institutions for inebriety and drug habits. Willmar costs the patient practically nothing, except a small per capita borne by the State. The average quack institution charges $150.00 for a cure, so called, whether the cure lasts for three days, or, as in some of the more conservative (?) quack institutions, the period is extended to ten days, and in the notoriously drink-habit cures, to thirty days. This ought to appeal to a doctor forcibly, inasmuch as all these claims of cures made by quack institutions are limited to thirty days at the outside, an absolutely absurd statement, and, for that reason, if for none other, we should all support and entertain anything that tends to increase the efficiency of the State farm for inebriates at Willmar.

One thing which Dr. Freeman wants to emphasize is the necessity at times of forcible restraint in a building especially constructed for detention cases. There is a small class of people who are, perhaps, suffering from a disease state, who are irresponsible. Most of them are common drunkards, who create all sorts of disturbances and who really need discipline—who need to be detained forcibly for a sufficient length of time to enable them to recover their normal physical tone, and until they recover something of their natural mental tone. If this could be incorporated in the rules and regulations of the governing body of the inebriate farm it would make a great increase in the total number of improvements and recoveries.

Dr. Freeman has emphasized the necessity of getting the physical condition up to a high point. He has said all that is really needed on the subject. I believe drugs and drink should be reduced rapidly in almost every case. If you look over some of the literature of some institutions that take these people, you will find they reduce the morphine down from fifty grains to forty, and then to thirty-nine, until, finally, after a period of so many weeks or months, they cut it down to the two-hundredth of a grain, and give it hypodermically. You can readily see the absurdity of that treatment. The average man can have the total reduction made within thirty-six or forty-eight hours.

I hope you will take more interest in the inebriate farm, and see that your legislator is interested as well.

Dr. Haldor Sneve (St. Paul): I have listened with a great deal of pleasure to Dr. Freeman's paper, and especially because there are some statistics as to what can be accomplished in such an institution even in a comparatively short time. Personally, I think that six months as an average time to stay in this institution would be too short. It will be found, however, in time, whether this is true, but just now the institution is in the experimental trial stage.

A great many legislators are, as Dr. Jones said, trying to convert this institution either into an insane asylum or a tuberculosis sanatorium; and it is up to the profession of the state to back up the establishment of this institution for the treatment of a class which is growing.

Personally, I think drink is a vice and not a disease, and until we can eradicate from the minds of the laity and from the minds of some physicians the idea that a man who drinks is some sort of a nervous invalid, the sooner we shall get better results in the handling of this question. Even the dipsomaniac has periodic brain-storms, which Dr. Ball has likened to attacks of migraine; that is a good simile, they do not always take to drink, but go off in other ways.

I have treated from twenty to fifty cases of delirium tremens at the City Hospital every year for twenty years, and I have had considerable experience in institutions; and yet I cannot find anything to criticize about the principles of treatment that Dr. Freeman has put forth here today. The idea in the minds of the laity is that inebriety is a disease, and they want drugs for it to make them well, and that is one reason why so many patients go to Keeley cures and get well. They go there because they find a drug that cures *disease*. I find that the Towne-Lambert treatment is an excellent *mental* treatment for the inebriate in private practice. It can be used in the institution at Willmar, as well as in private practice, and putting a patient upon the Towne-Lambert treatment satisfies his desire to cure the disease he is suffering from.

I think the profession will have to keep their eyes on the legislators, perhaps on the new governor, and see that this institution is not thrown into the waste-basket, so to speak, or converted into some other sort of institution, because we need a place of this kind. Even if Minnesota can go prohibition pretty soon—and I rather think it will—we shall not get rid of our drunkards for that reason. We shall still have to have a hospital for the treatment of the morphine, cocaine, and alcoholic habits. The doctors who send patients to Willmar, I think, ought to be careful, and not try to help some municipality out of taking care of old battered hulks, who cannot hope to recover, who cannot be made well simply because they have been drinking for so many years, and their other habits of life have resulted in such a deterioration of the brain that there is no possibility of bringing them back and making really good citizens of them. Those patients should be kept in a work-house or in a special department at Willmar or some other place. We should try to reclaim all of our young men and young women habitues.

Owing to the absence of proper writeups about this hospital it is not generally known throughout the state that pay-patients can be received and treated just as in any sanitarium and at very moderate rates.

Dr. Freeman (closing): I really have nothing to add in closing except to say a word with regard to prohibition. I have a second-hand statement from the police of one of the Twin Cities that he is positive in his city there are five thousand drug-users from his experience in the police court.

With regard to the maintenance of discipline at the institution: We have sufficient law or authority for discipline, but we have not the facilities. The thing in my opinion that we mostly require is a building where we can take care of a man who is incorrigible, or a man who runs away. For two reasons: In the first place, I have known a number of men who came there unwillingly, who later were greatly benefited by their compulsory stay; second, the effect of disciplinary measures upon the population in general. If a man knows that, when he goes there, he must stay, he naturally gets over his constant thought that he is going to sneak away, and put it over. The custodial cottage to take care of forty people would allow, in all, four classes of patients. We should have a reception-ward in which to examine all new patients; one ward for the incorrigible; and we should have two other places to care for two classes of men received. This would prevent the influence of the older men who have gone further in their habits upon the young boy who has just started.

DIAGNOSIS OF INTRACRANIAL COMPLICATIONS IN DISEASES OF THE MIDDLE EAR AND ACCESSORY SINUSES OF THE NOSE [3]

BY JOSEPH C. BECK, M. D.
CHICAGO.

[3] Read before the Sioux Valley Medical Association, July 22, 1914, and published in these columns at the request of the Association.

The most important causes of intracranial complication from the middle ear and nasal accessory sinuses, are suppurations, consequently I shall confine my remarks to that subject, and not take up the neoplasms, trauma, etc.

In the diagnosis it is most important to recognize suppurative disease of the ear and sinuses, but this subject is not within the province of this paper, therefore I shall satisfy myself by mentioning only that the presence of the pus from the middle ear and nose, and Röntgenographic examination, are the most important signs of affections of these structures. The one symptom more than any other on the part of the patient of a threatening extension into the cranial cavity, is localized pain or headache, which is very persistent, instead of periodic. Especially important is this in connection with the cessation or diminution of the discharge. The knowledge of the pathological change present in the sinuses and middle ear and mastoid, is of additional value as, for instance, tuberculosis, syphilis, and cholesteatoma.

The frequency of intracranial complication in suppuration of the middle ear is much greater than that following sinus disease, about twenty-five to one in my experience.

The intracranial complications which I shall consider are—

1. Meningitis.
2. Sinus thrombosis.
3. Brain abscess.

The meningitis may be serous or suppurative, and later localized or diffuse.

The sinus thrombosis may be partial or parietal, and complete with or without involvement of the jugular bulb and vein. The brain abscess may be extradural or genuine within the brain substance proper. The complications may be further divided as to bacteriologic or etiologic factors as, for instance—

Streptococcic
Staphylococcic
Pneumococcic
Tuberculous
Syphilitic

These complications may arise following acute, or chronic and acute, exacerbation of chronic suppuration of the ear and sinuses. Meningitis and sinus thrombosis (this latter condition is very frequently associated with a localized meningitis) are usually complications following acute, or acute exacerbation of chronic, suppuration of the ear and sinuses. Brain abscess, however, is most frequently associated with the chronic form of the ear and sinus disease; but these become more manifest following an acute attack of ear or sinus trouble. Tubercular or syphilitic meningitis is chronic inflammation *per se*; but these conditions are also lit up by the acute processes within the ear and sinuses.

The cardinal symptoms of any intracranial complications are—

1. *Pain or headache.*—This may be localized or diffuse; it is, however, very persistent and quite intense. It is in the recognition of this symptom that has helped me more than any other in suspecting intracranial trouble.

2. *Nausea and vomiting.*—This symptom is quite constant, especially early in the disease; and projectile vomiting is quite characteristic of intracranial pressure or irritation.

3. *General septic appearance.*—This of course will vary in the different conditions under consideration, but in all is it quite manifest.

4. *The vision* is very frequently affected due to the choked disk that is present.

5. *Temperature, pulse, and respiration* are very frequently disturbed.

6. *Definite focal symptoms* of brain localization are of the utmost importance in the diagnosis.

7. *Blood and spinal fluid examinations* give very valuable information.

8. *Röntgenographic findings* are at times valuable.

9. *Exploratory operation and treatment*, as in lues, is at times necessary to make a diagnosis.

MENINGITIS

(a) *Serous meningitis.*—One of the first signs is the increasing headache, at first localized, usually near the seat of the perforation or path of infection, and soon becoming diffuse over the head. The patient loses his appetite, his tongue becomes coated, the emunctaries become sluggish in their action, and nausea is a very common symptom. The temperature rises, and, if the septic form is going to follow, this rise is often quite rapid, so that there may occur small chills from the infection of the cerebrospinal fluid. The pulse and respiration rate is now considerably increased. The patient is very irritable and restless, and does not sleep. As soon as the fluid increases within the cavity there is observed the characteristic syndrome of rolling the eyes, especially upward, the neck is drawn backwards, and finally the leg upon the thigh and thigh upon the abdomen. Attempts to straighten them out is resisted and appears to be painful,—Kernig’s sign.

Stroking the bottom of the feet with some semisharp instrument or the finger-nail will cause the big toe to turn up instead of down,—Babinski’s sign.

Taking the head and tilting it forward against the chest will cause the limbs to be drawn up,—Brudzinski’s sign.

All the other symptoms, as pressing over the peroneal nerve and muscle (Gordon's sign), which will cause the extension of the toes, the stroking of the anterior tibial surface (Oppenheim's sign), or the stroking of the region of the external malleolus (Chaddock's sign), will produce retraction of the toes. All these signs, I say, prove that the upper neuron (within the cranium) is involved. The patient now will lapse into unconsciousness, and be roused with more or less difficulty to again relapse in the same condition. The pupils become sluggish in their action, at first becoming small, then irregular, and finally dilated.

Ophthalmoscopic examination may reveal a choked disk. Spinal puncture shows increased pressure by fluid very frequently coming through the hollow needle with a spurt, and clear or slightly cloudy. Following such a puncture the patient is very often much improved for from a half an hour to a whole day, but the symptoms soon return. A complete examination of the cerebrospinal fluid thus removed, will aid a great deal in diagnosis. This includes the following:

1. Remove about 25 c. c. at spinal puncture.
2. Make several slides and stains for organisms, as septic and tubercular.
3. Examine and count the endothelial cells, leucocytes, and pus cells.
4. Make cultures.
5. Make a Noguchi (butyric-acid) test for excess of albumin.
6. Make a Lange colloidial test.
7. Wassermann, Nonne, and Noguchi tests for syphilis.
8. Test for sugar.
9. Test for total acidity and relative acidity.
10. Cholin may be tested for.

In the serous form one will find the cells increased somewhat, especially the leucocytes, but the micro-organisms are conspicuous by their absence.

The Lange (colloidal-goldchloride) test will show the characteristic color reaction of a septic process.

The Noguchi (butyric-acid) test will be positive. Excess of albumin.

The Wassermann, Nonne and Noguchi tests for syphilis are negative. (Unless such a case should be a complicated one.)

The test for sugar is very important in that in serous meningitis sugar is present.

The relative acidity is not markedly affected, and cholin is not present, or, if so, in only small quantity.

(b) *Septic meningitis.*—If this is *localized*, and there is a collateral serous meningitis associated with it, then the symptoms may be the same, as just described; however, the cerebrospinal fluid will show a greater degree of irritation, and the fluid may contain some micro-organisms. The majority of localized septic meningitis cases, however, are not as severe in their course as the serous or diffuse septic forms. The one important symptom is the localized headache, which is quite persistent, and the greater rise in the temperature. There are, undoubtedly, many cases of localized meningitis that show a perfectly normal cerebrospinal fluid, and most of the cardinal symptoms absent; and these are the cases that usually get well or lead to extradural abscesses subsequently.

The *diffuse septic meningitis* is the most discouraging intracranial complication that we have to deal with, and the diagnosis as a rule is not difficult. It usually is preceded by the serous form, but within a very short time develops the graver symptoms of sepsis. The most positive symptom is the spinal puncture. The fluid comes out under pressure, but not so great as in the serous form, and is turbid. The turbidity varies in degree with the amount of infection. It has the appearance at times of pure pus; in fact, that is what it is. Bacteriologically one will find many micro-organisms of the character of the infection; and leucocytes or pus cells are very numerous.

The sugar reaction is always absent, and the acidity is much increased as is the quantity of cholin.

The pressure or irritative symptoms as the Kernig and Babinski tests, as well as the pupillary reactions, are practically the same as in the serous meningitis, only that they soon give away to the paralytic form, namely:

pupils dilate, patient is in a constant stupor or coma, and the involuntary urination and bowel movements become very manifest. The patient is, as a rule, unable to take or be given nourishment. The outcome is, in my experience, with one exception, always fatal, due to diffuse cerebritis. I have had a case of diffuse septic meningitis in the early stages of a pneumococcic type which I operated on by the Haynes' method of drainage of the cyscterna magna, and which recovered; and I believe that the success in that case was due to the very early intervention, because I have operated by the same method on eight other cases more advanced and of streptococcic and staphylococcic type of infection, which ended fatally.

Sinus thrombosis.—This complication is the one that is recognized as giving the best prognosis because it can be very readily recognized, and even exploration is warranted to make such diagnosis. It most frequently follows, or is associated with, acute infections of the middle ear and mastoid process. The most important symptoms are the chills and fever of a distinct septic type, and, as a rule, increasing in frequency. There is invariably a blood-picture of sepsis, namely, a very high leucocyte count and the polymorphonuclear type in marked excess. Blood cultures are, as a rule, positive of a bacteriemia. If the process has extended to the bulb and internal jugular vein, then one may feel a thickening or cord-like mass along the anterior border of the sterno-cleido-mastoid muscle. The fundus examination often reveals a choked disk, especially on the side where the thrombosis is located. A symptom recently described by Beck, of Vienna, and Crowe, of Baltimore, and proven by me to be of positive value in several cases, is the production or increase of a choked disk by compression of the healthy internal jugular vein. Urbanschitch has shown in quite a number of cases of sinus thrombosis that the blood-clotting time is very much enhanced. This of course is true of any case of bacteriemia or septic phlebitis anywhere in the body. I have proven this test to be of value to me in several cases of sinus thrombosis. The exploratory exposure of the lateral sinus is of distinct value, and the only fact to remember is to expose a sufficient area so that one is able to deal with the sinus in case it be opened accidentally, because such an accident when this precaution was not taken has led to serious consequences.

The diagnosis of a thrombotic sinus when exposed is made first by its discoloration, usually of a grayish pink; secondly, it feels harder than normal and is not resilient when compressed, that is, it does not spring back. It, however, may be soft in case the thrombus has broken down; and in cases of parietal thrombosis it may spring back because there is blood circulating through it. One will at times find a small collection of pus about the sinus, a condition known as perisinus abscess, and in many instances of this condition the sinus itself is not thrombosed. The puncture of the sinus by a hypodermic needle and attempt to withdraw some blood, is not at present considered good practice owing to the danger of infecting a non-infected sinus. An incision is considered a wiser plan, and subsequently packing both sides (torcular and bulb) so they are shut off from the general circulation. There are many instances of secondary infection by embolism, either in or about the joints, and infection into the lungs, spleen, pancreas, etc., with the entire train of symptoms from such complications.

Brain Abscess.—This is most frequently associated with chronic suppuration of the middle ear and mastoid, and labyrinthine disease. As stated before, we must consider two principal types, namely, those outside the dura and those within. They may exist at the same time, or the intradural abscess may frequently follow, especially in acute exacerbations, the extradural abscess. The paramount symptom is the great pain in the head, most frequently localized at or in close proximity to the abscess. I have, however, found several instances where the patient located the pain in the anterior portion of the head, and operation or post-mortem examination disclosed it in the posterior cerebral fossa. This pain is not at all unlike that in brain tumor, and there are exacerbations in the headaches sometimes at night, other times in the mornings, and in one of my cases the patient would have about ten attacks of severe head-pains within twenty-four hours, and in the intervals be fairly comfortable.

The next group of symptoms of importance are the focal lesions, which will correspond to the anatomicophysiologic locations and actions. These focal symptoms will vary in degree in that they be either irritative or destructive. So, for instance, a small abscess pressing over the motor area will cause clonic contraction and a still larger abscess, especially if it be intradural, will produce paralysis of that portion of the body governed by that particular area. Again, if it be located in the cerebellar region it will cause a train of symptoms of imbalance and loss of interpretation of direction, which must be carefully differentiated from the irritation of the labyrinth. In this department there has been much work done by Barany, Ruttin, Neumann, and other Viennese, and many others to make it possible to make a differential diagnosis; and there is a great deal more to be done. One of the most important recent contributions in this regard is the “pointing test” of Barany in connection with cerebellar lesions; and careful study and experimenting at every opportunity is very much recommended, in order to familiarize one’s self with this test. This in connection with the various labyrinth tests makes the differential diagnosis much more easy. One must remember that both labyrinthian irritation in connection with suppuration of the ear and cerebellar irritation from brain abscess may exist at the same time.

Intracranial pressure, being increased in brain abscess, will cause the cerebrospinal fluid to be increased and found to be so by spinal puncture, although no pus cells or micro-organisms will be found, unless there is also a concomitant diffuse septic meningitis or ventricular infection present. The ocular symptoms of intracranial pressure, such as pupillary (often one large and one small) and choked disk, are usually present. The *pulse rate* and *respiration* will be affected, as in brain tumor, according to the size of the abscess. The larger the abscess the slower the pulse and respiration. The temperature, as well as the pulse and respiration, will vary as to whether the abscess be intradural or extradural. Intradural abscesses will frequently cause considerable rise of temperature, and acceleration of the pulse and respiration, and a remission when the abscess has become partially walled off. As soon as a fresh invasion of brain tissue takes place another rise of temperature, etc., occurs.

Projectile vomiting is, as in brain tumor, quite frequently encountered.

The Röntgenogram, especially a stereoscopic one, will be of some value in cases where through its chronicity a change of bone by pressure has taken place, or if one may follow the path of necrosis from the nasal accessory sinuses or the middle ear and mastoid process towards the brain. I will state, however, as I have stated on several occasions before, that not too much emphasis should be laid on the diagnostic value of the *x*-ray in intracranial lesions, especially abscess. I have been disappointed in this great method of diagnosis (*x*-ray) and much annoyed at the positiveness of some observers without sufficient evidence.

As in sinus thrombosis, so in brain abscess one should not hesitate in the exploratory operation, because waiting too long will often reduce the patient's ability to stand an operation later on. Should one not find the abscess, then the decompression has done a great deal to prevent destruction of brain tissue by pressure, besides the patient will be very much relieved of the severe head-pains. This may be said also of spinal punctures. In this way one may wait for development of localization for another operation.

In conclusion, I would like to repeat the words of Prof. Neumann as to the differential diagnosis between meningitis, sinus thrombosis, and brain abscess: "A patient that has meningitis is one that wishes to be left alone and allowed to sleep, although when roused is not particularly irritable. If he has brain abscess then he is constantly very irritable and difficult to manage, while a patient that has sinus thrombosis when he is free from the chill and fever is very pleasant, apparently well."

THE TREATMENT OF GONORRHEAL OPHTHALMIA

Arthur Edward Smith, M. D.
MINNEAPOLIS

In ophthalmology, as in other branches of medical science, the advance in therapeutics has hardly kept pace, in recent years, with that in pathology and diagnosis. Comparatively few of the therapeutic innovations of the past decade have stood the test of time; and, in the main, the ophthalmological materia medica of today bears a striking resemblance to that of fifteen or twenty years ago. Our poverty of therapeutic resource has been notably exemplified in the generally accepted method of treatment of gonorrheal ophthalmia; and the results obtained with the conventional treatment as outlined in the current text-books are far from satisfactory.

Gonorrheal ophthalmia, in both infants and adults, continues to cause an appalling amount of blindness; and only a part of this can, with justice, be ascribed to ignorance and neglect. The number of cases which, in spite of the most careful treatment, go on to corneal ulcer, perforation, panophthalmitis, and irreparable blindness, continues to be considerable. Further, a decided difference of opinion still exists among well-trained oculists of wide experience as to the best method of handling these cases. For over a hundred years silver nitrate has enjoyed an unquestioned pre-eminence in the treatment of the purulent ophthalmias, particularly those cases in which the gonococcus was the etiological factor; and even now to question its right to a place in the treatment of gonorrheal conjunctivitis seems to many to be as heretical as to abandon mercury in the treatment of syphilis. For many years the only difference of opinion in regard to silver nitrate seemed to be as to whether it should be employed in the first stage of the disease, or whether one should wait until the discharge became purulent. Of late years, however, a number of experienced oculists have gone on record as being of the opinion that the majority of these cases do distinctly better without the nitrate than with it. As is well known, the nitrate destroys only those gonococci lying upon the surface or in the most superficial layers of the conjunctiva; and, far from reaching those in the deeper layers, rather forms a film over the surface which protects them from the irrigating solution used later. It also appears to be certain that the use of the nitrate, for a time at least, increases the ratio of extra-to intracellular gonococci in the discharge, which furnishes another valid argument against its use. That a subsequent chronic conjunctivitis with hypertrophy is often a disagreeable sequel in cases in which an energetic course of silver nitrate has been used is a matter of common observation. The vogue of certain of the organic silver salts, such as argyrol, protargol, etc., is no doubt, not so much due to any intrinsic therapeutic merit which they possess as to the fact that the average case gets along better without the local application of strong chemical antiseptics. However one may feel about the abandoning of such a time-honored drug as the nitrate of silver in the treatment of this disease, it must be conceded that it is entirely inadequate to control the process in the severer cases, and as a therapeutic sheet-anchor leaves a great deal to be desired.

The use of cold compresses in gonorrheal ophthalmia continues to be advocated in text-books and practiced in many clinics, especially in America, in spite of the fact that the progressive men in general medicine and surgery seem pretty generally to have abandoned the use of cold applications in the treatment of acute inflammations of bacterial origin. Any merit the cold compresses may have in the reducing of the edema and relieving pain are more than counterbalanced by the fact that the vitality of the tissues is at the same time lowered. In cases in which there is a sufficient swelling of the lids to cause a dangerous pressure on the eyeball, cold should not for a moment be depended upon to control the inflammatory edema but instant recourse had to canthotomy: in cases where this swelling is not marked cold compresses are unnecessary and apart from a certain analgesic effect, of no value. The skepticism, which is becoming more general, in regard to the value of silver nitrate and cold applications has not extended to the third member of the classic trinity,—irrigations,—the efficacy of which seems to be generally conceded. Various substances have been advocated for this purpose,—boric acid, potassium permanganate, bichloride of mercury, normal salt solution, etc., and the consensus of opinion seems to be that it is practically indifferent which one of these is used, the action being mechanical rather than chemical. The ordinary method of half-hourly irrigations has been abandoned by Hosford, Ulbrich, and others in favor of the constant irrigation with the Hosford apparatus or some modification of it.

The English adherents of the constant irrigation treatment, who, for the most part, dispense with the use of silver nitrate altogether, report excellent results; but the method is not without its drawbacks. The apparatus is awkward to use, requires as much or more attention than the intermittent irrigations, and undoubtedly disturbs the rest of the patient at night more. Further, since the lids are, of course, not held apart for the constant irrigation, but the flow of the solution across the palpebral fissure is depended on to cleanse the eye of secretion, one is inclined to question whether the mechanical cleansing is as thorough as when the lids are gently held apart while the eye is being irrigated.

The more one sees of these cases, the more one is impressed with two things: first, that a certain percentage of them would make a complete and uncomplicated recovery, even if they were entirely untreated (undoubtedly this number is larger than we think, especially in children); second, that the usual treatment is entirely inadequate in those cases in which there is an especially virulent infection or a lowered resistance of the tissues. When antigonococcic serum was first developed and its action observed in cases of acute gonorrheal ophthalmia, the results were, as in acute urethritis, disappointing. Many oculists are of the opinion that the serum is entirely without value in acute blenorrhea, even though its use be indicated in metastatic eye disease of gonorrheal origin. Of late, however, at least two men in America have written enthusiastically of serum-therapy in acute gonorrheal conjunctivitis, advocating its employment in the usual manner and also its use locally, i.e., dropped into the conjunctival sac in place of the usual antiseptics. It would seem that the data now available hardly warrant a positive statement in regard to the serum-therapy.

The pathological findings in gonorrheal ophthalmia are simple but significant, in that the gonococcus of Neisser is found, not only on the surface and in the superficial cells of the conjunctiva, but also, often within forty-eight hours, has invaded the deeper layers of the epithelium and the subepithelial connective tissue. This at once makes clear the reason for the inefficacy of the local antiseptics, particularly those like silver nitrate, the action of which is very superficial. Organic silver preparations and irrigations of various kinds are equally powerless to reach any but the most superficially situated of the bacteria.

Since the destruction of the bacteria lying on the surface is not sufficient to control the disease, it may be stated that the problem of the destruction or inhibition of the deep-lying bacteria is the essential problem in curing gonorrheal ophthalmia.

The gonococcus numbers among its biological peculiarities an unusual intolerance of extremes of temperature, its growth in culture being inhibited by temperature above 38° C. or below 18° C. Text-books on bacteriology state that exposure to a temperature of 60° C. for a period of ten minutes destroys the gonococcus. Experimentation in the laboratory of the Dimmer Clinic in Vienna in April and May, 1913, with cultures from forty-two cases of acute gonorrheal urethritis, seemed to indicate that this point may be placed from one and one-half to two degrees lower than this, i.e.,—from 58° C. to 58.5° C.

Thus, theoretically, at least, it would appear that, if the conjunctiva could be subjected to a temperature as near as possible to this without injury to the tissues, a marked effect should be observed in the course of the disease, particularly if the heat can be applied in such a way as to penetrate as deeply into the tissues as does the gonococcus. This theoretical requirement has, in my opinion, been perfectly met practically by the local use of steam as practiced in the Dimmer Clinic since February, 1913, with the apparatus devised by Lauber and modified by the writer. Goldzieher of Vienna was probably the first to employ steam in the treatment of the purulent ophthalmias; and in his first series of cases reported fifteen patients treated with the application of steam passing through a nozzle held at a distance of about four centimeters from the eye, the temperature of the steam striking the tissues being about 45° C. (113° F.). Although the results indicated that the method was a distinct step in advance there were still a number of important details to be worked out, in order to get the best possible results. First of all, experiment showed that the temperature of the steam at a given distance from the nozzle was not constant, so that an arbitrary distance could not be set. This suggested the advisability of providing the apparatus with a sliding-guard, which could be set at the exact distance from the nozzle where the steam was shown by the thermometer to be at the desired temperature. Secondly, it was determined that the tissues would sustain without injury a considerably higher temperature than that set by Goldzieher, and that the effect upon the diseased process was markedly better when the temperature was raised. Steam at from 50° C. to 53° C. gave the best results; and in one case in which a temperature of 55° C. was inadvertently reached no injury was done the tissues. Further experience naturally suggested other changes in the original technic. In the first place, the lids were held apart by an assistant in the usual manner; but, even with gloves on, the exposure of the fingers to the steam was more or less painful, and gauze wound on little sticks was substituted. The time of exposure was finally set at six minutes; and since the application of the steam could not be borne for longer than from forty-five to sixty seconds without severe pain it usually took twenty minutes or so to complete the six-minute exposure. This was done once every twenty-four hours, and was combined with half-hourly irrigations with potassium-permanganate solution. No other treatment was used. The results attained with this method in 34 cases (7 adults, 2 children and 25 infants) has left nothing to be desired. In no case has there been any corneal complication; swelling and pain subsided with unusual promptness; and the course of the disease was notably shortened, whereas, after the first application of silver nitrate a considerable increase in the number of gonococci in the discharge is often observed. A striking diminution in the number is noted after the initial application of the steam. In 8 of the cases in the series mentioned (all infants), the disease affected both eyes; and in 5 of these cases the experiment was made of treating one eye with steam in the manner described and the other with applications of silver nitrate in the usual manner, using the permanganate irrigations in both. The difference in the results attained was very striking. In every instance the eye in which the steam was used was brought much more quickly under control than the one under nitrate. In cases brought under treatment early the edema of the lids did not become severe; and the course of the disease seemed, in general, to be shortened by about one-third. There were no corneal complications, except in one case in which there was a corneal ulcer present when the man presented himself at the clinic. In no case was canthotomy necessary; and no case was followed by a chronic hypertrophic conjunctivitis. The application of the steam is undeniably painful, but not unbearably so.

VAGINAL HYSTERECTOMY UNDER SPINAL ANESTHESIA: REPORT ON A CASE

By R. R. Cranmer, M. D.
MINNEAPOLIS

I wish to report this case of vaginal hysterectomy under spinal anesthesia on a patient whose age and physical condition were not favorable for the use of ether or chloroform. The case belonged to that comparatively small class in which a general anesthetic cannot be used; and it was because of this fact that spinal anesthesia was resorted to. Had it not been necessary for this patient to earn a livelihood by hard labor the operation would not have been done; but, in her case, it was necessary, and the condition of prolapse, therefore, was a source of continual pain and trouble. The fact that the diet was not restricted after the operation assisted greatly in shortening her stay in bed and her rapid recovery.

> Patient, aged 59, married, mother of six children. She had been suffering from prolapsus uteri of a severe degree for five years. The cervix presented at the vaginal orifice at times. Mitral insufficiency and arteriosclerosis were present. She also had chronic bronchitis and a mild nephritis. Chloroform and ether being contra-indicated, spinal anesthesia was used, two drachms of 2 per cent novocaine solution being injected through the fourth lumbar interspace. The vagina was prepared for operation, and the hysterectomy started within four minutes after the spinal injection. The patient did not complain of any pain; and there was no shock or other untoward symptoms. She was immediately put upon a general diet and was able to leave the hospital on the twelfth day.

THE JOURNAL-LANCET

The Journal of the Minnesota State Medical Association
and Official Organ of the
North Dakota and South Dakota State Medical Associations

PUBLISHED TWICE A MONTH ESTABLISHED 1870

Subscription $2.00 a year

PUBLICATION OFFICE

839-840 Lumber Exchange Minneapolis, Minn.

March 1, 1915

A NEW REMEDY FOR PYORRHEA ALVEOLARIS

Diseased teeth and gums have an undoubted and pernicious effect upon the general health of the individual. This condition as a cause of disease has been the subject of many papers written by physicians and dentists.

Now a new remedy has been proposed by Bass and Johns which promises relief in the majority of cases. Emetin is the drug that destroys the ameba of pyorrhea just as ipecac destroys the ameba of dysentery. The lesion should be attacked persistently until healed and the use of emetin continued to prevent reinfection.

Emetin may be used hypodermically in one-half grain doses for at least three days and as often as is necessary to destroy the ameba.

The presence of the ameba can be determined only by proper microscopic examinations. The healing process may require considerable time, according to the extent and character of the necrosis. Deep pockets require careful cleansing to clear the pus-forming cavities. After this has been accomplished and pus ceases to form Bass and Johns recommend the use of fluid extract of ipecac as a local application to prevent reinfection. Ipecac will actually destroy the ameba if used persistently and is preferable to the many commercial preparations now in use. The teeth should be brushed in the ordinary way, after which one drop of fluid extract of ipecac should be applied to the wet brush, forcing some of the solution between the teeth and spitting out the excess without further washing of the mouth.

The investigators have found that this simple procedure will keep the mouth free from pyorrhea. It stands to reason, however, that the teeth must be thoroughly cleaned in the usual manner by the dentist, otherwise it will be impossible for the emetin or ipecac to penetrate the deep crusts which are found about old and uncared for mouths. It is remarkable how many people neglect the care of their teeth and it is equally strange that so little constitutional disorder is found in those who neglect an ordinary and simple sanitary toilet requisite. One of the first rules for hospital patients when they come under the supervision of the nurse is the provision of a tooth brush and a suitable mouth wash.

Many patients from the country, a lesser number from the cities, never employ a tooth brush. Some even resent a suggestion of clean teeth. Nature gave them teeth and nature is supposed to keep them in order, but unclean teeth are the rule rather than the exception in hospital practice.

Not infrequently animals need the services of a dentist, but their numbers are few compared to man. When a simple remedy for pyorrhea, like ipecac, promises to clear the teeth of amebas, there is no excuse for neglecting nature's adornment.

LOWERING THE MILK GRADE

A bill has been introduced in the Minnesota State Legislature for the purpose of lowering the butterfat requirement in milk from three and one-quarter to three per cent. This means a reduction of solids in milk from thirteen to eleven per cent, and it further means that more water will be added to much of the milk sold in Minnesota. A Minneapolis ordinance prescribes the butterfat content to be as high as three and one-half per cent. Minneapolis has enough water in its milk now, and, if this bill goes through, the city may expect to use skim-milk almost exclusively.

It hardly seems credible that any one should desire the quality of milk to be reduced for any purpose whatever unless it is for commercial reasons.

Fortunately, at this writing the bill is held up for consideration, and it is to be hoped that sufficient pressure will be brought to bear to insure its defeat. Too many cows give poor milk and any effort to standardize and legalize the inferior cow is a reflection on the integrity of milk sellers. Inferentially, there are too many under-fed children and yet if milk is reduced in quality, we must expect less vigor in the growing child.

One wonders why such a bill should get into the Legislature; what are the real reasons for its passage?

"LEAVES OF HEALING!"

The late issue of “Leaves of Healing,” published by the Dowieites at Zion City, near Chicago, has been sent broadcast among physicians. This sheet is an antivaccination propaganda, and is profusely illustrated by horrible pictures of supposed diseased states caused by vaccination. The text is, as is all others of its ilk, full of misinformation, garbled extracts from known and unknown writers and speakers, and tirades against all who believe in vaccination.

If these sheets would present a fair and broad view of the evils of vaccination they might find more adherents to antivaccination doctrines among medical men; but, as it contains so many misstatements and is so overbearingly one-sided in its efforts, the effect is nil, except when it is circulated among those unbalanced in mind and judgment. Physicians in general freely acknowledge that vaccination, or the introduction of a serum, may produce, in some people, unexpected and sometimes disastrous results. Most physicians hesitate to vaccinate people with active syphilis, or even those in whom the syphilis has been seemingly inactive for years, or those who have hereditary syphilis. These persons are quite apt to have an accentuation of their old blood disorder under slight infections or injuries; but that should not militate against vaccination when an epidemic is probable. Some of the pictures in “Leaves of Healing” were undoubtedly pictures of syphilis, and should have been so labeled; but that could not have been expected in a partisan publication.

Physicians also know that people who have chronic eczema should not be vaccinated until the eczema clears up; and doubtless in hurried or extensive vaccinations that are deemed necessary to prevent the spread of smallpox in a community cases of eczema are overlooked. Children who are the victims of chronic digestive disorders, or who react to mild febrile or diarrheal conditions more than the average child, are commonly exempted from vaccination. On the whole, there are but few conditions that are made worse by careful vaccinations with proper dressings and after-care.

When one considers what wonders in the way of control of smallpox have been recorded in medical history, the few mishaps that occur among the vaccinated, the proportion of illness due to vaccination is so infinitesimal that they cannot be classed among the “fearful” results of vaccination.

“Leaves of Healing” leaves out of its vaporings the fact that Zion City had a smallpox epidemic not long ago, and was quarantined by the health authorities, and that the people submitted to vaccination with gratifying results. Nor does the above-mentioned magazine record the fact that the president and secretary of a local branch of antivaccinationists in Minneapolis, who were fighting a compulsory vaccination law before the Minnesota Legislature a few years ago, died of virulent smallpox during that meeting of the Legislature.

The antivaccinationist usually has at his command a set form of speech that contains more vituperant adjectives, and less reason and judgment, than the average self-constituted reformer. Smallpox and other preventable diseases will continue to exist while the uneducated and ill-balanced minds are permitted their volley of wind-laden speech. Some day the people will wake up, cast the “reformer” aside, and climb on to the band-wagon of health and happiness.

It will take our educators and sanitarians some time to harness the team to the wagon, but when it starts it will go on merrily to its destination.

OWNERSHIP OF THE JOURNAL-LANCET

In answer to a number of inquiries the following statement is made:

The stock of the Journal-Lancet is held by a number of Twin City physicians, and the publisher, Mr. W. L. Klein.

The Journal-Lancet is the official organ of the State Medical Associations of Minnesota, North Dakota, and South Dakota. The responsibility for its reading matter and editorials rests with the publication committees of the state associations.

MISCELLANY

TO THE PHYSICIANS OF THE STATE OF MINNESOTA:

The Committee on Public Policy and Legislation most earnestly asks the co-operation of every physician in the State of Minnesota in procuring the passage of the several bills that have been decided upon, and either have been or will be introduced into the legislature during this session. It is believed that there is not a man upon the roster of the State Medical Society, or indeed any physician in Minnesota, who does not see the necessity of certain legislation for the protection of the physicians in the State, and also that the common weal will be advanced by the passage of the telephone bill introduced by Senator Andrews, of Blue Earth, and by the passage of the bill relative to trachoma, which is a constant menace to the public health, and several other bills that are in course of preparation, but which await certain developments before their presentation. The committee earnestly begs of all the physicians in the State that they will write to their representatives and senators from time to time urging with great earnestness their support for the several measures advanced by the Committee on Public Policy and Legislation. It is believed that every physician can influence at least from 10 to 100 votes at a general election, and this fact, of itself, makes the physician a factor in the election of any candidate. It is believed by this committee that the medical men of the State, if they will but unite and act in concert, can measurably influence legislation. The time has come for the physician to take his place in the political system of the State, both as an active agent and, indirectly, through his influence of others.

The telephone bill provides for physical connection between all telephone companies in the State without extra charge, except a small toll. It provides that telephone companies shall be placed under the direction of the Railroad and Warehouse Commission. It provides that no greater net income than 5 per cent shall be allowed upon the capital actually used in the operation of the telephone companies. It provides for intercity telephone service in the cities whose city limits adjoin without extra charge.

The trachoma bill provides for the segregation of trachomats, and, under certain circumstances, for the maintenance by the State of special schools for their education in school districts having as many as 20 trachomats.

There is also drafted and ready for introduction a bill requiring all persons who seek to practice medicine in any form whatever to pass the regular examination before the State Board of Medical Examiners.

There is in contemplation a bill for the purpose of procuring certain lands for the building of cottages thereupon and establishing farms to be worked by lepers who may be or shall have been committed to the leprosarium farm, the intention being that those lepers in the State that are able to work shall have an opportunity to do so, and that the said lepers should care for lepers who are unable to work or earn a living. It is also proposed to purchase a small tract of land not far from the State University for the purpose of allowing an exhaustive study of certain forms of leprosy with the aid of the State University Medical Staff. The leprosarium farm would be under the direction of the State Agricultural School.

The Chairman of this Committee will be very glad to receive advice and suggestions from the physicians in the State.

CORNELIUS WILLIAMS, M. D.,
Chairman of the
Committee on Public Policy and Legislation.
St. Paul, Minn., February 3, 1915.

REPORTS OF SOCIETIES

MINNESOTA ACADEMY OF MEDICINE

The Academy met at the St. Paul University Club, Feb. 3. Dr. C. M. Carlaw presided.

Four doctors were proposed for membership: Drs. W. H. Condit and Stephen Baxter, of Minneapolis, and Drs. Wilhelm Lerche and F. C. Schuldt, of St. Paul. All four names were referred to the executive committee.

Dr. Arnold Schwyzer showed some x-ray pictures of a penetrating gastric ulcer. He also made a report of a case where gall-stones gave a feeling of emphysematous crackling, due to small marble-sized stones with no more fluid than enough to fill the spaces between the stones (perhaps a teaspoonful in all).

The paper of the evening was presented by Dr. A. E. Benjamin, the subject being "Goiter Operations with Simplified Technic." The paper was thoroughly discussed, the whole evening being given over to its consideration.

The reading of Dr. White's thesis was deferred until another meeting.

Twenty-seven were present.

FRED E. LEAVITT, M. D., Secretary.

CORRESPONDENCE

To the Editor:

In the February 15th issue of The Journal-Lancet is a discussion by Dr. Klaveness, of Sioux Falls, S. D., on a paper on "Syphilis and Its Relation to Society" by Dr. McLaughlin, of Sioux City, Iowa. In this discussion Dr. Klaveness states: "We are unfortunate here in South Dakota in this respect, that we do not have the population and the laboratory facilities for resorting to the Wassermann reaction at all times, and any man within the State who would systematically carry out a Wassermann reaction now and then would invalidate his findings very materially, inasmuch as it is very well established that, in order to obtain reliable readings, you must have a serologist or bacteriologist to follow this work exclusively in order to get accurate findings. It is immensely important, and it would be a boon to the suffering people, if we could have a state serologist."

This statement by Dr. Klaveness is contrary to the facts as they now exist and did exist at the time he discussed the paper at Watertown, S. D., in May, 1914.

We have a well equipped medical laboratory in South Dakota in connection with the medical department at the State University at Vermillion, and we have been doing the Wassermann test.

This misstatement should have been corrected at the time it was made, but was not, as I was in Watertown but part of one day during the State Meeting last May and did not hear the paper or its discussion.

Permit me to state through your columns that we do the Wassermann test at the State Health Laboratory and have been doing it on Thursday of each week since March 21, 1914. At that time a circular letter announcing the fact was sent to every physician in the State, including Dr. Klaveness. This announcement was made only after several months of experimental work in perfecting the technic and controlling all factors.

We do the original Wassermann test, using the Nogouchi antigen. All our reagents are prepared in our laboratory and every possible control is carried out each time the test is set up. We therefore believe that our results will compare favorably with the best scientific work of this character.

At the present time a fee of $5.00 for each test is charged, containers and instructions are supplied upon request.

We have done the Wassermann test for the State Hospital for the Insane at Yankton from the first.

Mortimer Herzberg, M. D., Director.

Vermillion, S. D., February 18, 1915.

THE LOYALTY OF NURSES

To the Editor:

My attention has just been called to an article published in The Journal-Lancet, August 1, 1914, it being an address by Dr. George D. Head to the graduating class of the Asbury Hospital. The advice Dr. Head gives to the nurses seems very good, and very elevating to our profession, but I would like to analyze it to show that it is not quite practical.

It has taken considerable effort on the part of nurses to convince the people, and to convince some doctors, that they are any more than machines. Because we ask for three hours rest out of the twenty-four, and because we asked for a fixed rate for service, Dr. Head says that our loyalty to high ideals is diminishing. Unfortunately, in the nursing profession, as in all other professions, there are some who are incapable and unconscientious; and, if Dr. Head had the experience of having a nurse leave a patient, unattended, at a critical time, she probably was one of the few incapables, or was so overtired from loss of sleep that it was necessary for her to have rest. When Dr. Head says that a nurse should waive her rest hours for days or a week at a time, if necessary, I think he is making a mistake. A nurse cannot do her duty by a patient if she does not have proper rest. It is unfair to both the patient and the nurse. Dr. Head may say that most patients are not in need of constant attention for more than a few days or a week, and that a nurse can stand it for that length of time without rest hours. This is true; but we have to consider that the next case may be just as critical, and so the nurse must reserve some strength for the cases to follow. And more often than not, the nurse is obliged to take cases with very little or no rest between them.

In the second place, Dr. Head thinks that the nurses ought to have a varying scale of charges for service. The doctors do it; why shouldn't the nurses? Dr. Head does not seem to consider the fact that the nurse has one patient, while the doctor has many. Suppose a nurse takes care of a poor patient for five or ten dollars a week, where is the rich patient who is willing to pay forty or fifty dollars a week to make up the loss? The nurses have found that twenty-five dollars a week is the price that is necessary for them to live on in order to keep themselves clothed, pay for their laundry (no small item), and carry them over the few weeks of rest or over the dull season. The average life of a nurse, as a nurse, is, I believe, not more than ten years. In that length of time, at the wages she gets, she is not able to lay away a great amount for a rainy day, which usually comes all too soon.

We have a number of good hospitals in Minneapolis where people in moderate circumstances can be very comfortably cared for at a considerably less expense than employing a nurse in their homes. The poor in our city, I think, are fairly well taken care of in the city hospitals and by the visiting nurses, who are paid for such work.

As for nurses refusing cases because they are afraid of them: I think there is usually some just cause. If a nurse has a tendency towards tuberculosis, she should refuse such cases; or if she has a tendency towards throat troubles, she should refuse diphtheria and scarlet-fever cases. A nurse who is constantly with a patient runs considerably more risk of infection than the physician, whose visits are usually short. There are nurses who make a specialty of such cases, and usually there is no trouble finding such a nurse. Nurses who make a specialty of obstetrical cases or of children should not take contagious work. As for a nurse refusing a typhoid case because she is afraid of it: I cannot believe that any real nurse would do such a thing.

It also seems to me very ridiculous, and it surely cannot be a common thing for a nurse to inquire before she consents to take a case whether or not the plumbing is modern and how many servants are kept.

As to just what Dr. Head means by saying that a nurse should be willing to do any kind of service about a house. I do not know; but I do know that nurses are not usually physically fit for washing or scrubbing, yet, as a rule, nurses are glad to perform duties which are not just in their line, in order to help the household to run smoothly.

Most of the nurses in general work are engaged in nursing because they are obliged to earn their living, and in most cases because they are especially interested in this particular field; and, although most nurses take some charity cases, it is impossible for them to take many, even to satisfy what Dr. Head calls "the inner, higher longings of the soul."

HARRIET M. PRIME, R. N.

Minneapolis, February 4, 1915.

BOOK NOTICES

MANUAL OF OBSTETRICS. By Edward P. Davis, A. M., M. D., Professor of Obstetrics in the Jefferson Medical College, Philadelphia. 12mo of 463 pages, 171 illustrations. Philadelphia and London: W. B. Saunders Company, 1914. Cloth, $2.25 net.

As indicated by the name this is a handy book. It is well illustrated, the text is brief and well written, and as complete as could be expected in a work of its size.

It presents no features which are especially new, though it takes up many of the most recent advances in obstetrics.

It is a work that aims to give those who wish it a concise account of the status of obstetrics at the present time.

—ADAIR.

BALNEO-GYMNASTIC TREATMENT OF CHRONIC DISEASES OF THE HEART. By Prof. Dr. Theodor Schott, Bad-Nauheim. Published by Blakiston, Philadelphia. Price, $2.50.

This brochure sets forth in the main, preceded by a short chapter on medical treatment, the philosophy, technic, and clinical results of balneogymnastic therapy in chronic heart-conditions.

It would appear, inasmuch as Prof. Schott admits the non-establishment of the probable curative factors of either the carbon dioxide or mineral constituents, that possibly, as Dr. Anders in the foreword surmises, the curative properties may reside in the "advantage of being far removed from the cares and responsibilities growing out of the practical affairs of life at home."

Relative to the more firmly established value of the gymnastic element, it is quite evident that the "resistance movements," are an improvement over the Zander mechanico-gymnastic, and of similar value to the so-called "Terrain Kur," with the added advantage of personal application.

—Schneider.

Diseases of Bones and Joints. By Leonard W. Ely. M. D., 220 pages, 94 illustrations. Surgery Publishing Co., N. Y. Price, cloth. $2.00.

Few men are better fitted than Dr. Ely to write an authoritative book on joint and bone diseases. He has gone at his study from the only logical end; that is, the study of the underlying pathology. The book throughout shows the result of much conscientious work in the pathological and x-ray laboratories, carefully checked up from the clinical aspect.

The average specialist who writes a manual for the use of the general practitioner seems to think he must mention every theory which has ever been brought out since the time of Hippocrates, together with a list of every form of treatment ever proposed. The bewildered family doctor gets about as much help as he would from the perusal of a few pages of the Index Medicus. One good theory, clearly stated, even if it is not universally accepted, may form a practical working basis which will be of great aid to the doctor in the understanding and care of his cases. In this particular Dr. Ely is most satisfactory. He has worked out the pathology of the tubercular and other chronic joint diseases in a clear and logical manner. While much may have to be altered in the light of further research, at least one can feel sure that Dr. Ely has convictions, and that his work will form a useful basis for further investigations.

The illustrations are taken almost entirely from photographs or photomicrographs and are mostly original. So many works on Orthopedic Surgery appear which are filled with cuts handed down from one text-book to another, cuts of impossible people wearing impossible appliances, that it is hard to imagine that such a book, illustrated with such quaint old prints, can represent the latest word or offer anything new.

Dr. Ely's discussion of the pathology of joint tuberculosis is perhaps the most interesting thing in the book. His idea of the red bone marrow and the synovia being the sole tissues to be primarily involved does not agree with the recent work of Fraser, of Edinburgh. Evidently more work must be done in order to harmonize these findings.

In general, while not much space is given to treatment, what there is, is clear and is carefully selected by the author, instead of leaving this important point to the discretion of the reader.

His discussion of the chronic arthritides is quite full and very instructive. He points out particularly the resemblance between the pathological conditions found in various chronic infectious joints and in the various stages of joint tuberculosis.

This little book will certainly be of use to anyone who has to treat bone and joint diseases.

—Reed.

NEWS ITEMS

Dr. Chas. Pierce, of Wadena, has moved to Menahaga.

Dr. J. L. Stewart, of Spearfish, S. D., has located at Custer, S. D.

Dr. Jas. Farrage, formerly of Deering, N. D., has located at Park Rapids.

Dr. Hugo Neukamp is leaving Fessenden, N. D., to locate in Beulah, N. D.

The Dell Rapids Hospital was completed and opened the latter part of February.

Dr. H. A. Gueffroy, of Chicago, has taken over the practice of Dr. D. F. Sullivan, of Frankfort, S. D.

The new St. Alexius hospital, at Bismarck, N. D., was formally opened to the public February 15th.

The entire surplus of the old Homeopathic Hospital Association, amounting to $1,000, was voted to the support of the Maternity hospital, of Minneapolis, at a recent meeting.

The Physicians' Hospital company has been incorporated at Thief River Falls for the purpose of building and maintaining a hospital at that place. The company is capitalized at $25,000.

In a previous issue we stated that Dr. G. P. Shepard, of Chicago, had located at Jamestown, N. D. Dr. Shepard is from Courtenay, N. D., and not from Chicago, though he has been taking postgraduate work in that city for the past few weeks.

The Medical Society of the State of New York invites all physicians of the country to its hundred and ninth annual meeting which is to be held in Buffalo, April 27-29. This will probably be the largest medical meeting of the year, except perhaps that of the A. M. A. in San Francisco.

Messrs. J. D. Edgar, Arnold Hamel, R. A. Johnson, and H. A. Rudd, and Miss Olga Hansen, all of the class of 1915, have been elected to the

Minnesota chapter of Alpha Omega Alpha, the national honorary fraternity in medicine, the membership of which is based solely upon scholarship.

Dr. James E. Moore, who has practised in Minneapolis for thirty-two years, twenty-eight of which have been devoted to the exclusive practice of surgery, has given up his practice and will, hereafter, give his entire time to the Medical School of the University of Minnesota, except for a limited amount of consultation work.

In our last issue we made the statement that the Ramsey County Medical Society would not admit a physician to membership until he had been a resident of the county for one year. This should have been written so as to convey the meaning that a physician must have been a resident of some county for at least a year, not necessarily Ramsey County.

It is the desire of the publishers of the Journal-Lancet to make this department of news as interesting to its readers as possible. The items are obtained from a number of sources, and, though a great deal of care is given to their preparation, mistakes will necessarily occur. Will you not help to keep up the interest of this column by sending in anything which may be of interest to the readers? Notify us of mistakes as they occur that we may make a correction in the next issue.

"The Mayo Foundation for Medical Education and Research, Incorporated," with an initial endowment fund of $1,500,000, has recently been incorporated. It has for its object the endowment of the graduate medical instruction and research work which has for years been a feature of the Mayo Clinic, at Rochester. The founders are: William J. Mayo, Charles H. Mayo, Henry S. Plummer, Edward Starr Judd and Donald C. Balfour. The board of temporary trustees having in charge for the present the investment of the fund is composed of Bert W. Eaton, George W. Granger and Harry J. Harwick. The board of scientific directors is composed of Louis B. Wilson, William F. Braasch, E. Hessel Beckman, A. H. Sanford, and Walter D. Sheldon. For the present the expenses of the foundation will be met by annual contributions from the Mayo Clinic, the income from the endowment being allowed to accumulate and increase the principal.

PHYSICIAN WANTED

To locate in a thriving North Dakota town. For full information correspond with Andrew Erickson, Makote, N. D.

an A 1 physician and surgeon with a good stand in or very near the Twin Cities, with the view of effecting an exchange for part of the year. Address, 204, care of this office.

FOR SALE

To a man with surgical ability, one-half interest in my private practice and well-equipped hospital, located in a live up-to-date county-seat town in Minnesota; population 2,500; two railroads; good schools and roads; good fees. This is an excellent opportunity to get into a place with a good future. Price, $5,000 for one-half interest in hospital building, equipment, office fixtures, and practice; $2,500 cash. Don’t write unless you mean business and have the cash. Address 202 care of this office.

Doctor: If you want practical post-graduate work during the fine season in the delightful city, write for particulars. Twenty-eighth annual session opens September 28, 1914, and closes June 5, 1915. New Orleans Polyclinic, P. O. Drawer 261, Post-graduate Medical Dept., Tulane University of Louisiana.

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Diabetes, though not always curable, is controllable. Practically all diabetics can be made sugar-free and the acidosis disappears with the sugar. By a special regimen the reappearance of the sugar and the acidosis may be prevented.

The Battle Creek method is based upon experience gained in the treatment of many hundreds of cases supplemented by the observations and discoveries of Von Noorden, Falta, Guelpa, Benedict, Allen, and numerous other investigators. The essential features of the method are—

1. A thorough preliminary examination and repeated examinations comprising (a) complete quantitative examination of the urine daily, (b) differential study of the blood, (c) chemical, microscopic and bacteriological examination of the feces and study of the pancreatic function, (d) X-ray examination of the stomach and intestine with special reference to stasis.

2. Study of the patient's metabolism by the respiration apparatus to determine his respiratory quotient, CO_2 tension and basal ratio.

3. Establishment, by the aid of metabolism studies of each case, of a regimen adapted to the individual by determining the proper proportion of protein, fats and carbohydrates to keep the urine free from sugar. The kind of protein, fat and carbohydrate is considered important, as well as the amount.

4. The patient's metabolism is regulated by baths, voluntary and automatic exercise, photo- and thermotherapy and other physiologic means.

5. The results of the regimen and treatment are accurately controlled by a "Metabolism Graphic" which shows the daily variations in the amount of urine, amount of sugar, acidosis, coefficient of sugar utilization, coefficient of carbohydrate utilization, nitrogen balance, glucose nitrogen ratio, weight balance and energy balance. These factors are all worked out by expert chemists and dietitians and with this data before him, and a great variety of

special foods of known energy value suited to diabetics at ready command, and the assistance of a strong corps of specially trained dietitians, the physician is able easily to arrange a dietary adapted to each case and to note each patient’s progress with the most careful scrutiny.

Under this comprehensive management the sugar usually disappears from the urine in two or three days, and does not return so long as the prescribed regimen is followed.

A few weeks’ treatment usually suffices to train the patient to a suitable dietary which he may safely follow under the guidance of his home physician.

We will be glad to send full information concerning the Battle Creek Method in Diabetes to any physician who will mail to us the attached coupon.

The Battle Creek Sanitarium, Battle Creek, Mich.

Box 350
The SANITARIUM,
Battle Creek,
Michigan

Please send to the undersigned full information concerning the Battle Creek method of treating diabetes.

Dr..............................

Street..........................

City............................

State...........................

PUBLISHER'S DEPARTMENT

QUAKER OATS

Doctors all over the country are advising the use of oat foods for the old as well as the very young child. It is for young folks developing, for grown-ups, who are hard workers, and for the old folks who wish to keep young. You can safely use Quaker Oats. It costs no extra price, and when you use it you are certainly getting the very best in oat foods. Physicians should read their page announcement in this paper.

ELECTRO-THERAPY

The Scheidel-Western X-Ray Coil Co. announce on another page that they have the latest appliance in electro-therapeutics, namely, the Columbia Treatment Transformer No. 9.

The apparatus is illustrated and described on another page, and more fully in the Company's new catalog.

As the Company is the largest manufacturer of x-ray apparatus in the world, their catalog should be in the hands of every man using this line of treatment.

BOREMETINE—A NEW EMETINE PREPARATION FOR PYORRHEA

Every doctor and dentist in the United States should know about this new preparation for the local treatment of pyorrhea alveolaris. Boremetine is a 1-2 per cent solution of emetine hydrochloride, together with boric acid, zinc sulphocarbolate, and aromatics.

The emetine is amebicidal, the boric acid bactericidal, and the zinc sulphocarbolate astringent. These three drugs meet the three essential factors necessary for the successful treatment of pyorrhea. Boremetine should be used in every case, either alone, or (in some severe cases) in association with the subcutaneous injection of Emetine Hydrochloride (Abbott).

A special free booklet on “Pyorrhea Alveolaris: How to treat it successfully with Emetine” will be sent on request. Send for it today. The Abbott Alkaloidal Company, Chicago.

OCONOMOWOC HEALTH RESORT

The State of Wisconsin has an enviable reputation for not a few things in which it excels all the other western or middle-western states. One of these is its sanatoriums, or health resorts. The climate, the beautiful scenery, the pure water, the proximity to the large cities of Milwaukee and Chicago, and the high-grade medical specialists made it possible to found such institutions in southern Wisconsin long before other parts of the middle west had the population or transportation facilities to make success in this line either probable or possible.

The Oconomowoc Health Resort is one of the best equipped and best managed of these institutions. It accepts only nervous and mild mental cases. It is under the management of Dr. Arthur W. Rogers as resident physician. Dr. Rogers has both the professional equipment and the personality that are necessary in the treatment and care of persons suffering from nervous and mental disorders.

BATTLE CREEK SANITARIUM

Fifty years ago examination was largely a matter of pulse finding; now it is possible to weigh and measure the organic functions of the body with as much accuracy as is possible in the testing of an intricate mechanism. This becomes possible through a series of tests in many of which elaborate equipment is required. Perhaps no other institution has a more complete organization for diagnosis than the Battle Creek Sanitarium.

The physical inventory possible there is a very thorough and accurate stock taking of the vital functions. Many business and professional men visit the sanitarium each year in order to take full advantage of the diagnostic facilities.

An interesting booklet, “The Measure of a Man,” is offered free by the sanitarium to those who care to know more regarding the system of examination.

ARMOUR & COMPANY

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It is a pure preparation.

It is free from objectionable chemicals.

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Pituitary Liquid is required in such particular classes of cases that the practitioner can afford to use the best only.

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We shall be pleased to send you a sample of Pituitary Liquid with literature.

Note the name of the perfect pituitary preparation—*Pituitary Liquid (Armour)*.

THE DELICATE SCHOOL GIRL

Even the most robust and generally healthy children show the deleterious results of the modern system of educational “forcing” that prevails in most of our larger cities. The child that starts the school year in excellent physical condition, after the freedom and fresh air of the summer vacation, in many instances, becomes nervous, fidgety, and more or less anemic, as the term progresses, as the combined result of mental strain and physical confinement in overheated, poorly ventilated school-rooms. How much more likely is such a result in the case of the delicate, high-strung, sensitively organized, adolescent girl? It is certainly a great mistake to allow such a girl to continue under high mental pressure, at the expense of her physical health and well-being, and every available means should be resorted to to conserve the vitality and prevent a nervous breakdown. Regularity of meals, plenty of sleep, out-of-door exercise without fatigue, open windows at night and plenty of nutritious food, should all be supplied. Just as soon as an anemic pallor is noticeable, it is a good plan to order Pepto-Mangan (Gude) for a week or two, or as long as necessary to bring about an improvement in the blood state, and a restoration of color to the skin and visible mucous membranes. This efficient hematinic is especially serviceable in such cases, because it does not in the least interfere with the digestion nor induce a constipated habit.

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A number of typographical errors have been corrected silently.

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The two lines below were swapped from the original:
by a chronic hypertrophic conjunctivitis. The
canthotomy necessary; and no case was followed

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